

UNIT-5 $\Rightarrow$ Logistic Regression

27-10-25
Monday

* It comes under "prescriptive Data Analytics"

* Here we prescript the values (decⁿ making)

* Linear regresⁿ

$\downarrow$
relⁿ b/w dep & indep variable.

$\rightarrow$ outcome is y: con-
tinuous form

$\rightarrow$ used for predicting
the next values.

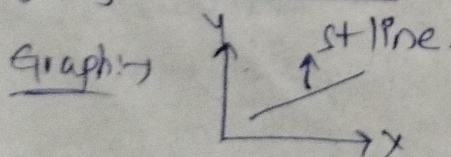
$$\rightarrow f^n: y_i = \alpha + \beta x_i + \epsilon$$

$\downarrow$
continuous value

$\rightarrow$ First we need to
find y_i & convert
it into prob.

$\rightarrow$ Here we use
loss fn $\Rightarrow$ Mean sq^d error

$\rightarrow$ Main aim is to
minimize error by
using M.S. Error



Logistic regⁿ

$\downarrow$
It is also relⁿship b/w
I/p & o/p variable.

$\rightarrow$ outcome is y:

in the form of category

$\Rightarrow$ probability $\Rightarrow$ classification

$\Rightarrow$ used for classification

$\Rightarrow$ The outcome is like a
category type.

$\rightarrow$ Sigmoid fn converts
continuous value into probab-
-ility

$$\sigma(y) = \frac{1}{1 + e^{-y}}$$

euler's No.
 $e = 2.72$

$\rightarrow$ It converts continuous
value into prob. value.

$\rightarrow$ Range: 0 to 1

$\rightarrow$ After we need to
describe it is belong to
which class.

$\rightarrow$ Here we have cutoff
value i.e. "Threshold value"
Decision boundary

i.e. 0.5

* Based on this value,
we can classify / categorize
the value. $\rightarrow$

* After converting continuous value into prob, based on threshold it will be classified

* If $prob < 0.5 \rightarrow 0$ (belongs to 0 class)
 * If $prob \geq 0.5 \rightarrow 1$ (belongs to 1 class)

* Here we use loss fn $\rightarrow$ 'cross entropy' which is used to 'maximizing the likelihood'.

* for binary classificⁿ ($M=2$)

cross entropy fn $= - (y \log(p) + (1-y) \log(1-p))$
 (no of classes)

* for ($M > 2$) : (multiple classifⁿ)

cross entropy fn $\rightarrow - \sum_{c=1}^M y_{o,c} \cdot \log(p_{o,c})$

Where M = no. of classes

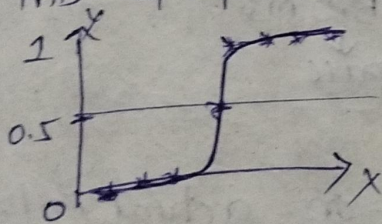
y = Actual class

p = predicted class

c = correct class [$= 1$]

o = otherwise [$= 0$].

* In this Graph is $\rightarrow$ "S" shape



Ex: A grp of 20 students studied b/w 0 & 6 hrs for an exam. we want to predict the prob of passing based on study hrs.

Study hrs	0.5	0.75	1	2	3	4.5	5.5
y values (P/F)	0	0	0	0	1	1	1

Q81: First we need to do linear regression & need to find y value

$$y_i = \alpha + \beta x_i + \epsilon$$

Then we get α & β . $\alpha = -0.2757$
 $\beta = 0.2566$

$$y_i = -0.2757 + 0.2566 x_i$$

$$\text{If } x_i = [5, 0.75, 5.5]$$

$$y_i = [1.0073, -0.0833, 1.1357]$$

* Logistic regression: $\sigma(y) = \frac{1}{1 + e^{-y}}$

$$\text{If } y = 1.0073 \Rightarrow \sigma(y) = \frac{1}{1 + e^{-1.0073}}$$

$$\text{prob value} \leftarrow \sigma(y) = 0.73$$

$\Rightarrow \text{prob} [0.73, 0.48, 0.76] \Rightarrow \text{[for 3 values of } y]$

↓ 1 (pass)	↓ 0 (fail)	↓ 1 (pass)
By spending 5 hrs we can have chance to pass	45 min ↓ fail	5.5 hrs ↓ pass

* Confusion Matrix

* It is a table based on actual values & predicted values.

* From this, we can calculate accuracy, precision, Recall/sensitivity, specificity, F1-score

* Predicted values

	actual values	
True	True	False
True	TP	FN
False	FP	TN

True → From actual
True, how many are identified
True - True

Accuracy ^{metrics} $\rightarrow$ It is used to measure overall correctness. How many pred's were right out of all pred's.
 i.e. ratio b/w true pred's out of all pred's.
 (correct)

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

* precision $\rightarrow$ out of all predicted +ve's how many were actually +ve.

* It is a true prediction.

$$\text{Precision} = \frac{TP}{TP + FP} \quad [\text{vertical}] //$$

* Recall / Sensitivity / True +ve Rate $\rightarrow$

* out of all ^{actual} +ve's how many were correctly predicted. ~~actual~~

$$\text{Recall} = \frac{TP}{TP + FN} \quad [\text{Horizontal}]$$

* Specificity / True -ve Rate $\rightarrow$

* out of all ^{actual} -ve's how many were correctly predicted as -ve.

$$\text{Specificity} = \frac{TN}{FN + FP}$$

* F1-score $\rightarrow$ It will balance the precision & recall when classes are imbalanced.

$$F1\text{-score} = 2 * \left(\frac{\text{Precision} * \text{Recall}}{\text{precision} + \text{Recall}} \right)$$

Q.4 In covid test of 1000 patients. there were 45 +ve tests of which 30 patients had covid & 15 were falsely tested positive of the 955 -ve tests. there were 5 that were incorrect, draw the confusion matrix & find the accuracy, precⁿ, Recall, specificity & F1-score.

Soln: 1000 $\rightarrow$ 45 +ve $\rightarrow$ 30 TP, 15 $\rightarrow$ FP
 955 $\rightarrow$ -ve $\rightarrow$ 950 TN, 5 $\rightarrow$ FN

		Actual		
		+ve	-ve	
pred	+ve	TP 30	FP 15	Total = 1000
	-ve	FN 5	TN 950	

$$i) \text{ Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{30 + 950}{30 + 15 + 5 + 950} = \frac{980}{1000} = \boxed{0.98}$$

$$ii) \text{ prec}^n \Rightarrow \frac{TP}{TP + FP} = \frac{30}{45} = \boxed{0.666}$$

$$iii) \text{ Recall} \rightarrow \frac{TP}{TP + FN} = \frac{30}{35} = \boxed{0.857}$$

$$iv) \text{ Specificity} = \frac{TN}{FP + TN} = \frac{950}{950 + 15} = \boxed{0.984}$$

$$v) \text{ F1-score} = 2 \left(\frac{\text{prec}^n * \text{Recall}}{\text{prec}^n + \text{Recall}} \right)$$

$$= 2 \left(\frac{0.666 * 0.857}{0.666 + 0.857} \right)$$

$$= 2 \left(\frac{0.666 * 0.857}{1.523} \right) = \boxed{0.749}$$

Q. Given a sample of 12 individuals, 8 that have been diagnosed with cancer & 4 that are cancer free where individuals with cancer belongs to class 1 (+ve) & Non-cancer belongs to class 0 (-ve). We can display that data as follows:->

Individual values	1	2	3	4	5	6	7	8	9	10	11	12
Actual value	1	1	1	1	1	1	1	1	0	0	0	0
Predicted value	0	0	1	1	1	1	1	1	1	0	0	0

Q.14 total actual +ve $\rightarrow 8$.

TP $\rightarrow 6$ FP $\rightarrow 1$.

Total -ve $\rightarrow 4$.

TN $\rightarrow 3$ FN $\rightarrow 2$.
Actual.

		Actual	
		+ve	-ve
Predicted value	+ve	TP. 6.	FP. 1
	-ve	FN. 2	TN. 3.

$$\text{Accuracy} = \frac{9}{12} = \boxed{0.75} //$$

$$\text{precision} = \frac{6}{7} = \boxed{0.857} //$$

$$\text{Recall} \rightarrow \frac{6}{8} = \boxed{0.75} //$$

$$\text{Specificity} \rightarrow \frac{3}{4} = \boxed{0.75} //$$

$$\text{F1-score} = 2 * \left(\frac{0.857 * 0.75}{0.857 + 0.75} \right) = \boxed{0.807} //$$

(* Regression Evaluation Metrics

* Used for minimizing the error.

- 1) MAE → Mean Absolute Error.
- * The avg of absolute diff b/w predictive & actual values.

$$MAE = \frac{\sum_{i=1}^n |y_i - \hat{y}|}{n}$$

- 2) MAPE → Mean Absolute percentage error.
- * The avg of absolute percentage error.

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}}{y_i} \right| * 100$$

- 3) MSE → Mean squared Error.

* The avg of squared diff b/w predictive & actual values.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y})^2$$

- 4) RMSE → Root mean squared error.

* The sq. rt of MSE.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y})^2}$$

* These metrics are used for regression & forecasting.

* Forecasting Techniques

* Based on past data, it will predict the future value is called "forecasting".

* In this we have forecasting process.

* Forecasting Process

1. Analysis $\rightarrow$

* It will take historical data & identify particular components in data.

Time series $\rightarrow$ means seq of obsⁿs recorded over time with equal intervals.

Time series components $\rightarrow$

i) Trend (T) $\rightarrow$ It takes long period of time to inc/dec a sales (cost of sales)

$\rightarrow$ This comp is used when we want long period of time calcuⁿ.

Ex: Gold rate.

* It records census every 10 yrs

ii) Seasonal (S) $\rightarrow$

* It is a short period of time inc/decⁿ of sales.

* It will calculate season wise.

* period time — below 1 yr

iii) Cyclical (C) $\rightarrow$

* It is a long period — more than 1 yr.

* Every year, at that time, the same thing is repeating.

* Every year it is repeating.

* period time $\rightarrow$ more than 1 yr.

iv) Irregular (I)

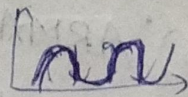
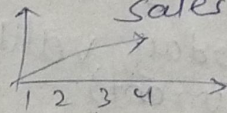
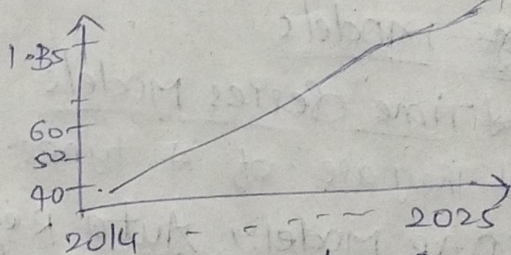
There are some other factors which influence sales (wars --)

The sudden changes / random changes

we can't predict the time $\rightarrow$ unpredictable time.

At these times, the sales are influenced.

Some natural phenomena / calamities (Tsunami, etc.) will influence the sales.



- * sudden/random changes influence the sales
- * we can't predict the future values.
- * we can predict future values of Trend, Seasonal, cyclical.

* In this analysis, it will take the historical data to predict the future values.

* Based on historical data, it will find the component.

Models

Time Series Models

* They are of 4 types.

(1) AR Model: Auto Regressive model.

* It is indicated with "(P)"

* today value is depends on previous result (current)

(2) MA(q): Moving Avg model

* current value is depends on previous error

(3) ARMA(P&q): Auto regressive moving Avg model

* current value depends on previous result & previous error.

~~(4) ARIMA(P,d,q)~~

* These 3 all belongs to stationary
i.e. data is stationary (There is no diff)

* Here we have const mean & variance.

* There is only small diff.

(4) ARIMA(P,d,q): Auto regressive Integrated moving average.

* It is a non-stationary data.

↓ It converts non-stationary data to stationary data & after that ~~we~~ apply models.

* Based on time series :-

Year	Price
2014	40K
2015	50K
2016	60K
!	!
2025	1.4L

↓ ARIMA will differentiate it.

↓ ~~TS~~ diff = 1 [1 yr diff]

↓ After finding diff, if data is not stationary, then takes TS = 2. [2 yr diff]

↓ Differentiate with 2 yrs.

↓ Like this we convert non-st data to stationary data & apply ARMA model to predict the future value.

Formulas :-

AR:
$$y_t = \beta_0 + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \dots + \beta_n y_{t-n}$$

time series

MA:
$$y_t = \epsilon_t + \beta_1 \epsilon_{t-1} + \dots + \beta_n \epsilon_{t-n}$$

ARMA:
$$y_t = \beta_0 + \beta_1 y_{t-1} + \dots + \beta_n y_{t-n} + \epsilon_t + \beta_1 \epsilon_{t-1} + \dots + \beta_n \epsilon_{t-n}$$

* In ARIMA, the differentiated is done automatically.

* Prediction

* After applying models we need to predict the future values.